

Speech in NLP: Speech-to-Text (STT / ASR)

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1. Introduction: ASR goals, history, and the arc from classical to deep learning
2. Classical ASR: noisy channel model, WER, and HMM–GMM architecture
3. Acoustic features and training: MFCCs, mel filterbanks, GMMs, and Viterbi alignment
4. Hybrid HMM–DNN ASR: neural acoustic modeling and recurrent hybrids
5. End-to-end ASR: Connectionist Temporal Classification (CTC) and RNN-Transducer (RNN-T)
6. Self-supervised learning: Wav2Vec 2.0
7. Modern ASR: Conformer, SpecAugment, and Whisper
8. Streaming ASR: causal and chunked encoders, RNN-T, and endpointing
9. Deployment, fairness, benchmarks, and references

- ▶ **Broad goal.** Given audio, produce exact transcript of words from speaker(s).
- ▶ **Verbatim (Exact) Transcription:** Capturing every “um,” word fragment, repetition, and disfluency to preserve the true speech signal.
- ▶ **Clean (Readable) Transcripts:** Text that omits or normalizes minor speech errors and filler words, yielding more fluent, reader-friendly output.
- ▶ **Real-Time, Low-Latency Operation:** For interactive settings (voice assistants, live captions), ASR must process audio as it arrives. Keep end-to-end latency under 200 ms for natural dialogue flow.
- ▶ **On-Device and Edge Constraints:** Need to run ASR on devices with no cloud support. Compute and memory may be limited.

Switchboard corpus conversation example

► In 2010:

- ASR, TTS, dialog all used specialized, hard-to-build modeling approaches
- Industry application of SLU systems limited. ASR “didn’t quite work well enough”

► Today:

- ASR, TTS, dialog all use deep learning approaches. Less specialized and better performance
- Spoken language systems are everywhere!
- New tools enable building full systems

Automaton:

- ▶ Markov 1906 (chains), 1913 (letter sequences in *Eugene Onegin*)
- ▶ Turing 1936
- ▶ McCulloch-Pitts neuron (1943)
- ▶ Shannon (1948) link between automata and Markov models

Human speech processing

- ▶ Fletcher at Bell Labs (1920's)

Probabilistic/Information-theoretic models

- ▶ Shannon (1948)



Radio Rex toy and box (c. 1920s)

Source: Stanford CS224S, Lecture 9 (2025 Spring),
slide 46.

1920's Radio Rex

- ▶ Celluloid dog with iron base held within house by electromagnet against force of spring
- ▶ Current to magnet flowed through bridge which was sensitive to energy at 500 Hz
- ▶ 500 Hz energy caused bridge to vibrate, interrupting current, making dog spring forward
- ▶ The sound “e” (ARPAbet [eh]) in Rex has 500 Hz component
- ▶ Any sound with enough energy near 500 Hz sets it off, so this is energy detection rather than word recognition

- ▶ **1952: Bell Labs single-speaker digit recognizer**
 - Measured energy from two bands (formants)
 - Built with analog electrical components
 - 2% error rate for single speaker, isolated digits
- ▶ **1958:** Dudley & Balashek built classifier that used continuous spectrum rather than just formants
- ▶ **1959:** Fry & Denes (UCL) ASR combining phoneme-sequence statistics and acoustic probability

► Hidden Markov Model 1972

- Independent application of Baker (CMU) and Jelinek/Bahl/Mercer lab (IBM) following work of Baum and colleagues at IDA

► ARPA project 1971–1976

- 5-year speech understanding project: 1000 word vocab, continuous speech, multi-speaker
- SDC, CMU, BBN
- Only 1 CMU system achieved goal

► 1980's +

- Annual ARPA “Bakeoffs”
- Large corpus collection:
 - TIMIT
 - Resource Management
 - Wall Street Journal

Hub5'00 Evaluation (Switchboard / CallHome)

(Possibly trained on more data than SWB, but test set = full Hub5'00)

WER (SWB)	WER (CH)	Paper	Published
4.9%	9.5%	An investigation of phone-based subword units for end-to-end speech recognition	April 2020
5.0%	9.1%	The CAPIO 2017 Conversational Speech Recognition System	December 2017
5.1%	9.9%	Language Modeling with Highway LSTM	September 2017
5.1%		The Microsoft 2017 Conversational Speech Recognition System	August 2017

WSJ

(Possibly trained on more data than WSJ.)

WER eval'92	WER eval'93	Paper	Published
5.03%	8.08%	Humans Deep Speech 2: End-to-End Speech Recognition in English and Mandarin	December 2015
2.9%		End-to-end Speech Recognition Using Lattice-Free MMI	September 2018
3.10%		Deep Speech 2: End-to-End Speech Recognition in English and Mandarin	December 2015

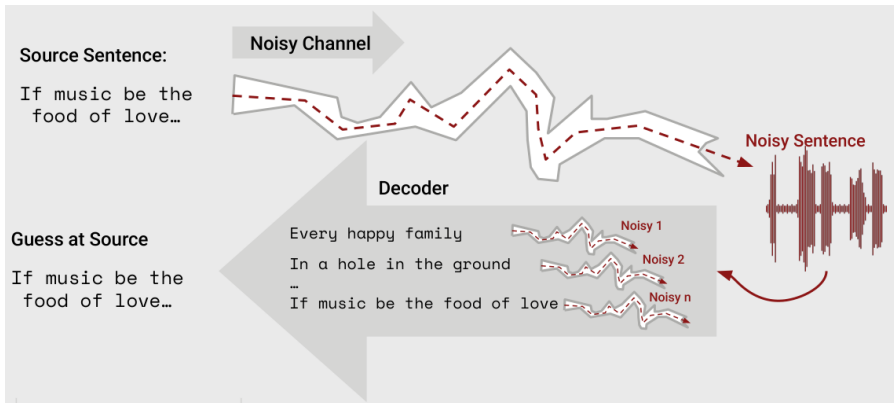
LibriSpeech

(Possibly trained on more data than LibriSpeech.)

WER test-clean	WER test-other	Paper	Published
5.83%	12.69%	Humans Deep Speech 2: End-to-End Speech Recognition in English and Mandarin	December 2015
1.9%	3.9%	Conformer: Convolution-augmented Transformer for Speech Recognition	May 2020
1.9%	4.1%	ContextNet: Improving Convolutional Neural Networks for Automatic Speech Recognition with Global Context	May 2020

Source: Excerpt of the WER progress tables at github.com/syhw/wer_are_we (Synnaeve); via Stanford CS224S, Lecture 9 (2025 Spring), slide 52.

- ▶ Search through space of all possible sentences.
- ▶ Pick the one that is most probable given the waveform.



Source: Stanford CS224S, Lecture 9 (2025 Spring), slide 5.

- ▶ What is the most likely sentence out of all sentences in the language L that generated some given **acoustic input** O ?
- ▶ Treat acoustic input O as sequence of individual observations:

$$O = o_1, o_2, \dots, o_T$$

- ▶ Define a sentence as a sequence of words:

$$W = w_1, w_2, \dots, w_n$$

- **Probabilistic implication:** Pick the highest probability word sequence W :

$$\hat{W} = \arg \max_{W \in L} P(W | O)$$

- We can use Bayes' rule to rewrite this:

$$\hat{W} = \arg \max_{W \in L} \frac{P(O | W)P(W)}{P(O)}$$

- Since denominator is the same for each candidate sentence W , we can ignore it for the argmax:

$$\hat{W} = \arg \max_{W \in L} P(O | W)P(W)$$

Likelihood



Prior



$$\hat{W} = \arg \max_{W \in L} P(O|W)P(W)$$

Source: Stanford CS224S, Lecture 9 (2025 Spring), slide 9.

$$\text{Word Error Rate} = 100 \times \frac{\text{Insertions} + \text{Substitutions} + \text{Deletions}}{\text{Total Words in Correct Transcript}}$$

Can be >100%.

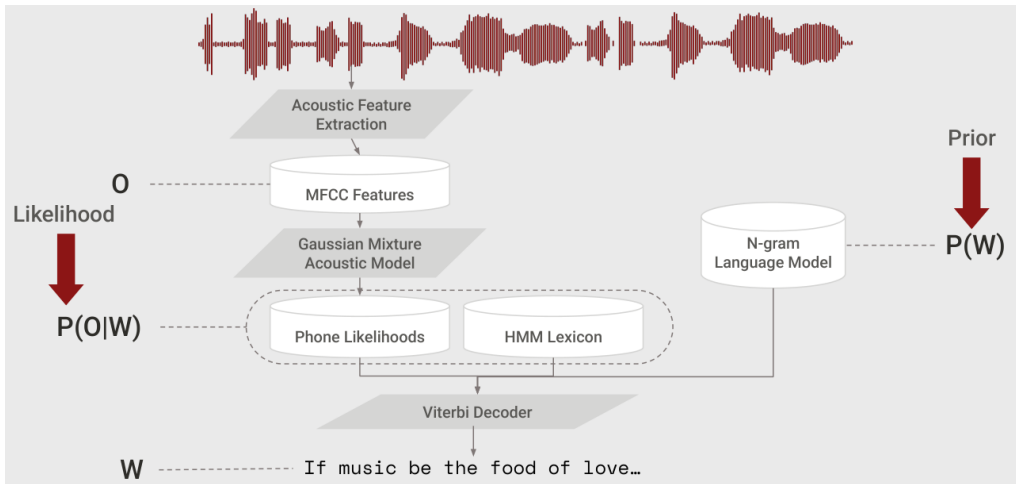
Doesn't distinguish between function words (of, they, he, she) and more important content words

Compute best alignment of reference and hypothesis to count errors:

REF:	i	***	**	UM	the	PHONE	IS		i	LEFT	THE	portable	****	PHONE	UPSTAIRS	last	night
HYP:	i	GOT	IT	TO	the	*****	FULLEST	i	LOVE	TO	portable	FORM	OF		STORES	last	night
Eval:	I	I	S		D	S		S	S			I	S	S			

Source: Alignment example after Jurafsky & Martin, *Speech and Language Processing*; via Stanford CS224S, Lecture 9 (2025 Spring), slide 11.

Speech Recognition Architecture



Source: Stanford CS224S, Lecture 9 (2025 Spring), slide 13.

- ▶ Words are too coarse to model acoustics directly, so the classical pipeline routes through a phoneme sequence L :

$$P(O | W) \approx \sum_L P(O | L) P(L | W)$$

(the sum rule over pronunciations, then the conditional-independence assumption $P(O | L, W) \approx P(O | L)$: given the phonemes, the audio no longer cares which words they spell).

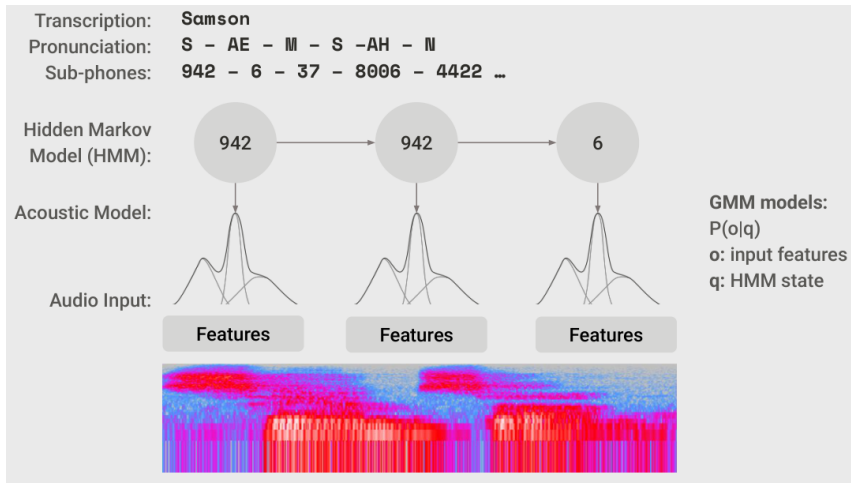
- ▶ This is exactly the architecture on the previous slide, as three learned objects:
 - $P(O | L)$ — the **acoustic model**: HMM states with GMM or DNN emissions (next frames);
 - $P(L | W)$ — the **lexicon**: a pronunciation dictionary (CMUdict-style); probability 1 for a single pronunciation, $1/J$ when a word has J of them, and a G2P model for out-of-vocabulary words — the same front end the TTS lecture teaches;
 - $P(W)$ — the **language model** (next frame).
- ▶ Every module is trained and tuned separately — the strength, and the curse, of classical ASR.

- ▶ The acoustic model and lexicon cannot separate “go to”, “go two”, and “go too” — all three produce the same phoneme string G OW T UW. Only $P(W)$ can.
- ▶ Counting in the WSJ text corpus (6.4M sentences):

Sequence	Count	Example
“go to”	2,710	the default reading
“go two”	2	“...often go two hundred miles or more”
“go too”	41	“he could go too far”

- ▶ An n -gram model stores exactly these statistics — the counting machinery from the first NLP lecture; neural language models (RNNLMs then, any LLM now) replace counts with learned context.
- ▶ The LM is the pipeline’s language-dependent text-side module — and it trains on *text alone*, no audio required. Remember that property for the next section.

Classical ASR: HMM-GMM system



Source: Stanford CS224S, Lecture 9 (2025 Spring), slide 16.

► **Markov assumption:**

$$P(q_t \mid q_1, \dots, q_{t-1}) = P(q_t \mid q_{t-1})$$

► **Output-independence assumption:**

$$P(o_t \mid O_1^{t-1}, q_1^t) = P(o_t \mid q_t)$$

$Q = q_1, \dots, q_N$ a set of N states
 $A = a_{ij}$: **transition probability matrix** A , each a_{ij} representing the probability of moving from state i to state j . $\sum_j a_{ij} = 1 \ \forall i$
 $O = o_1, \dots, o_T$ a sequence of T observations, each one drawn from a vocabulary $V = v_1, \dots, v_W$
 $B = b_j(o_t)$ a sequence of **observation likelihoods**, also called *emission probabilities*, each expressing the probability of an observation o_t being generated from a state j
 q_0, q_F a special **start state** and (**final**) **end state** that are not associated with observations, together with transition probabilities $a_{q_0 i}$ out of the start state and $a_{j q_F}$ into the end state

► Feature Extraction:

- 39 MFCC features

► Acoustic Model:

- Gaussian mixture models (GMM) to approximate $p(o|q)$

► Lexicon/Pronunciation Model:

- HMM: which phones can follow each other. Sequence constraints from phonetic lexicon

► Language Model:

- N-grams for computing $p(w_i|w_{i-1})$

► Decoder:

- Viterbi algorithm: dynamic programming for combining all pieces to estimate a word sequence from audio

Source: Stanford CS224S, Lecture 9 (2025 Spring), slide 15 (adapted).

You are a climatologist in the year 2799 studying global warming trends

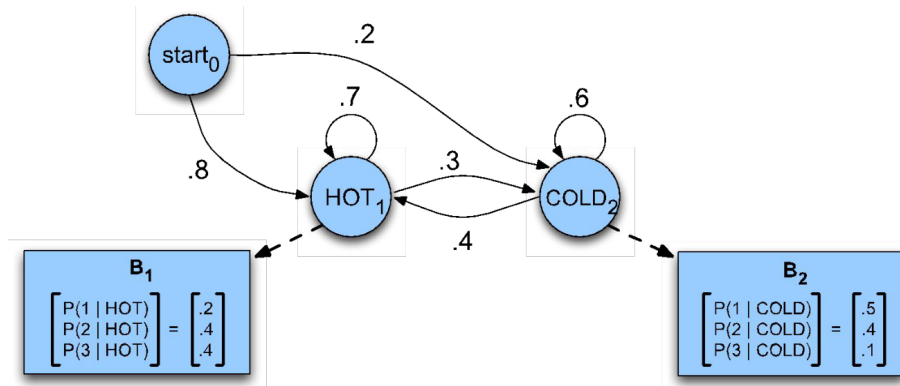
- ▶ You can't find any records of the weather in Baltimore, MD for summer of 2008, but you find Jason Eisner's diary! Which lists how many ice-creams Jason ate every date that summer

Our job: Infer the sequence of daily temperatures (HOT vs COLD)

- We do not observe temperature. We observe number of ice creams eaten
- There is a higher probability of eating more ice cream on hot vs cold days
- There is some relationship of next day temperature given current day temperature

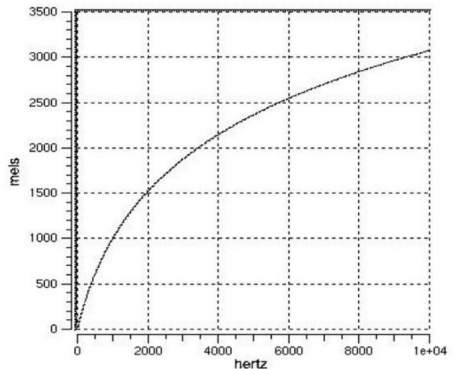
Simple HMM: Estimating climate from ice cream observations

Estimate the unobserved state daily_temp (HOT, COLD) based on observations of ice creams consumed



Source: Jason Eisner's ice cream task, after Jurafsky & Martin, *Speech and Language Processing*; via Stanford CS224S, Lecture 9 (2025 Spring), slide 19.

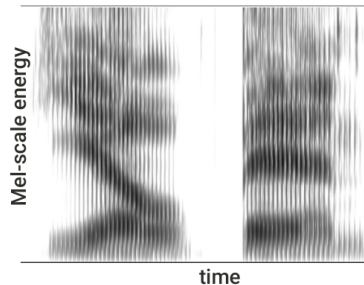
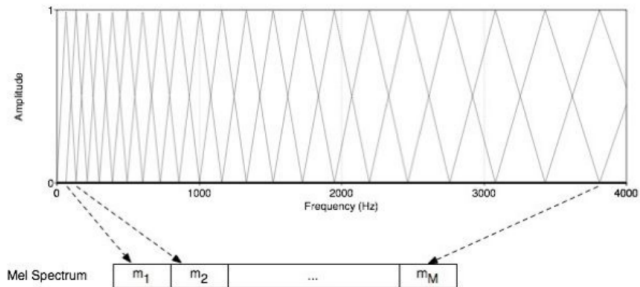
- ▶ Human hearing is not equally sensitive to all frequency bands
- ▶ Less sensitive at higher frequencies, >1000 Hz
- ▶ I.e. human perception of frequency is non-linear
- ▶ The **mel scale** makes equal steps in mels correspond to equal steps in perceived pitch



Source: Stanford CS224S, Lecture 9 (2025 Spring), slide 22.

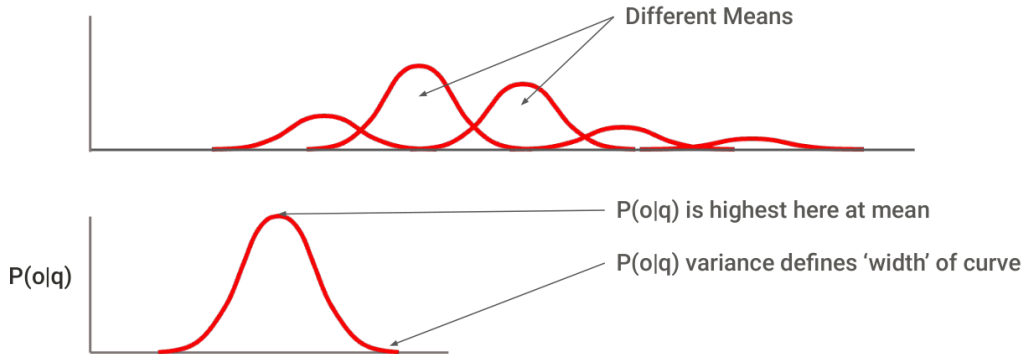
- **Mel Filter bank**

- Roughly uniformly spaced before 1 kHz
- logarithmic scale after 1 kHz



Source: Stanford CS224S, Lecture 9 (2025 Spring), slide 23.

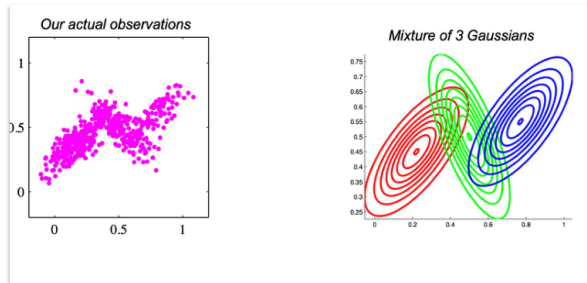
- $P(o|q)$: A single Gaussian parameterized by mean and variance. Mixture is a weighted sum.



Source: Stanford CS224S, Lecture 9 (2025 Spring), slide 27.

Summary: each state has a likelihood function parameterized by:

- M Mixture weights
- M Mean Vectors of dimensionality D
- either:
 - M Covariance Matrices of $D \times D$
- or more likely:
 - M Diagonal Covariance Matrices of $D \times D$ equivalent to:
 M Variance Vectors of dimensionality D



Source: Mixture-of-Gaussians panel © 2021 Carlos Guestrin, Stanford CS229; via Stanford CS224S, Lecture 9 (2025 Spring), slide 28.

- ▶ **Given our lexicon + HMM structure, and some acoustic model, we can:**
 - Generate the best alignment of HMM states to acoustic observations
- ▶ **With an alignment of HMM states to observations:**
 - Build a new acoustic model. Treat current state/obs mapping as training data+labels
 - This acoustic model is hopefully better than previous one
- ▶ **Repeat the align → rebuild acoustic model process until convergence**
 - Add parameters / complexity to acoustic model each iteration

Hybrid HMM-DNN ASR

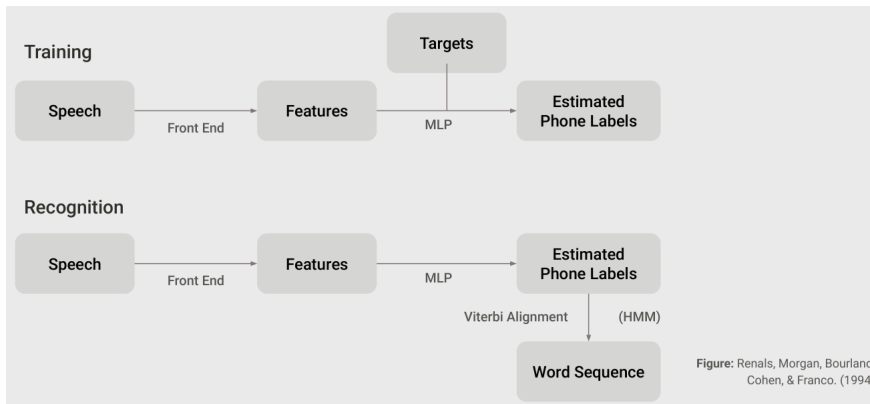
Transition to deep learning in acoustic modeling:
Hybrid HMM-DNN systems

- ▶ **Supervised learning**, minimize our classification errors
- ▶ **Standard choice: Cross entropy loss function**
 - Straightforward extension of logistic loss for binary

$$\text{Loss}(x, y; W, b) = - \sum_{k=1}^K \mathbb{I}(y = k) \log f(x)_k$$

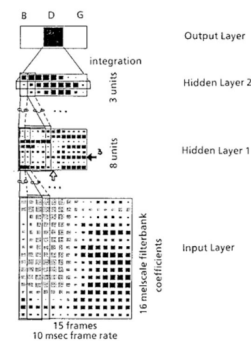
- ▶ This is a **frame-wise** loss. We use a label for each frame from a forced alignment
- ▶ Other loss functions possible. Can get deeper integration with the HMM or word error rate

Neural nets for acoustic modeling goes back to early speech

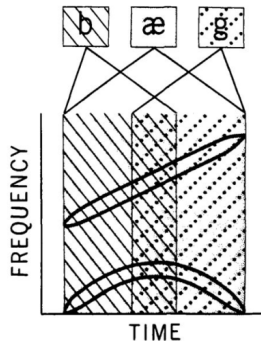


Source: Renals, Morgan, Bourlard, Cohen & Franco, "Connectionist Probability Estimators in HMM Speech Recognition," IEEE Trans. Speech and Audio Processing, 1994; via Stanford CS224S, Lecture 10 (2025 Spring), slide 10.

Speech tasks motivated some of the original backpropagation work



Waibel, Hanazawa, Hinton, Shikano, & Lang. 1988



McClelland, & Elman. 1985

Source: Left: Waibel, Hanazawa, Hinton, Shikano & Lang, "Phoneme Recognition Using Time-Delay Neural Networks," IEEE Trans. ASSP, 1989. Right: McClelland & Elman, "The TRACE Model of Speech Perception," Cognitive Psychology, 1986. Via Stanford CS224S, Lecture 10 (2025 Spring), slide 11.

Hybrid systems took over ASR around 2012-2015

- ▶ DNN-HMM hybrids (Hinton et al., 2012) rapidly replaced GMM-HMM on large-vocabulary tasks.
- ▶ Sets the stage for deeper encoders, sequence training, and later end-to-end models.

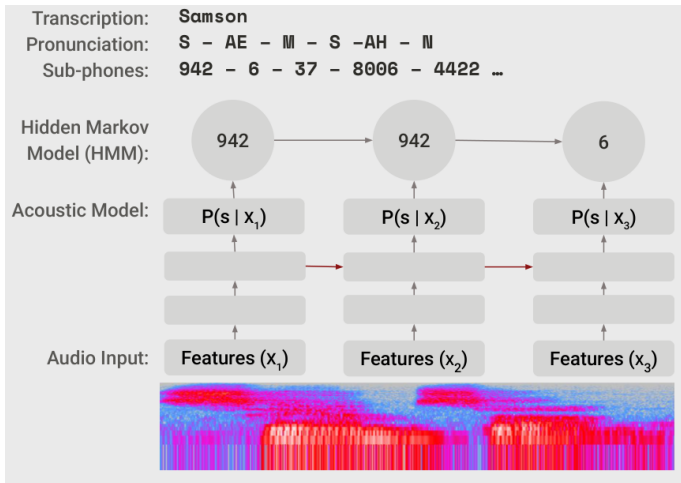
[TABLE 3] A COMPARISON OF THE PERCENTAGE WERs USING DNN-HMMs AND GMM-HMMs ON FIVE DIFFERENT LARGE VOCABULARY TASKS.

TASK	HOURS OF TRAINING DATA	DNN-HMM	GMM-HMM WITH SAME DATA	GMM-HMM WITH MORE DATA
SWITCHBOARD (TEST SET 1)	309	18.5	27.4	18.6 (2,000 H)
SWITCHBOARD (TEST SET 2)	309	16.1	23.6	17.1 (2,000 H)
ENGLISH BROADCAST NEWS	50	17.5	18.8	
BING VOICE SEARCH (SENTENCE ERROR RATES)	24	30.4	36.2	
GOOGLE VOICE INPUT	5,870	12.3		16.0 (>> 5,870 H)
YOUTUBE	1,400	47.6	52.3	

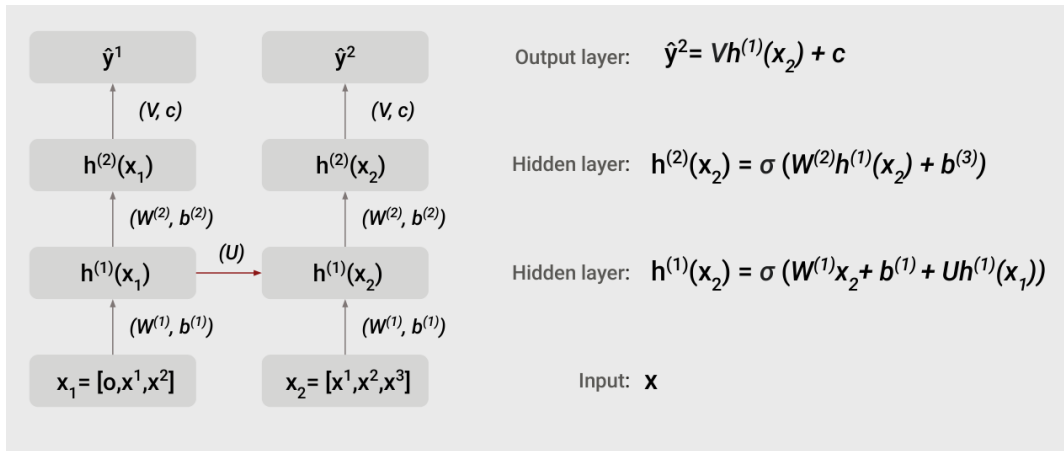
Source: Hinton et al., “Deep Neural Networks for Acoustic Modeling in Speech Recognition,” IEEE Signal Processing Magazine 29(6), 2012, Table 3.

- ▶ Context-dependent HMM states
- ▶ Deeper nets improve on single hidden layer nets
- ▶ Hidden unit nonlinearity
- ▶ Many more model parameters (scaling up) & large datasets
- ▶ Specific depth (e.g. 3 vs 7 hidden layers)
- ▶ Fast computers = run many experiments
- ▶ Architecture choices (easiest is replacing sigmoid)
- ▶ Pre-training **does not matter***
 - Initially we thought this was the new trick that made things work

Recurrent DNN hybrid acoustic models



Source: Stanford CS224S, Lecture 10 (2025 Spring), slide 15.



Source: Stanford CS224S, Lecture 10 (2025 Spring), slide 16.

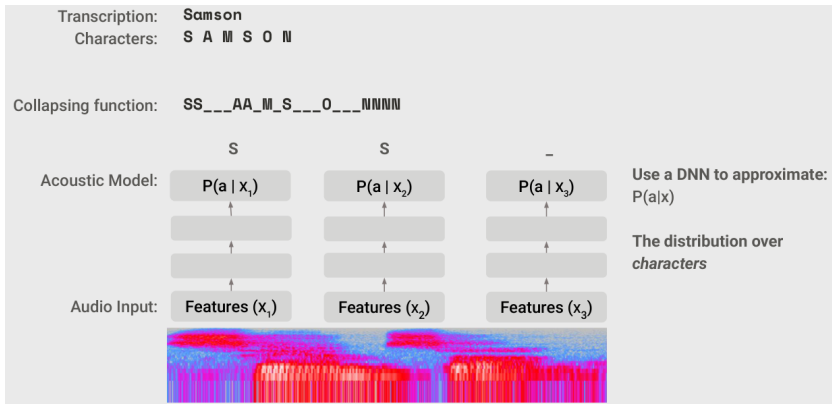
- ▶ Lexicon introduces too many pronunciation assumptions. Requires hand engineering.
- ▶ Iteratively building HMM systems requires complex “recipes” to progressively improve alignments + acoustic models.
- ▶ HMM-DNN systems perform better, but still require the above
 - Can we use deep learning approaches to replace the HMM-based approaches so far?

End-to-End ASR

An alternative to hybrid HMM-DNN: map audio directly to transcript tokens (characters, phonemes, BPE)

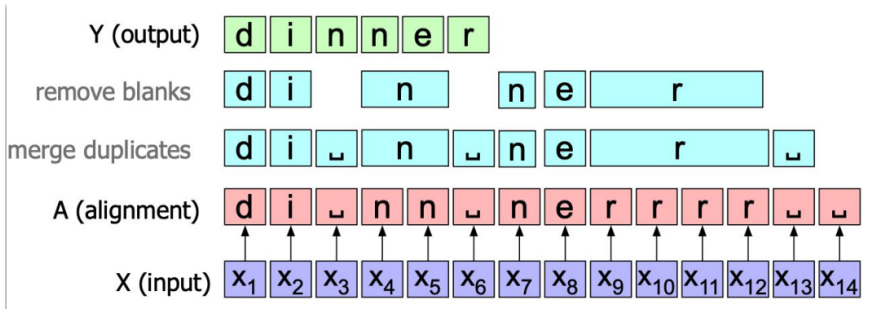
- ▶ **Modularity was a real asset.** Porting a classical system to a new language means swapping the lexicon (a dictionary) and retraining the language model (text-only data); the acoustic model, mapping audio to phonemes, is largely language-independent.
- ▶ **End-to-end models give that up:** mapping audio straight to text couples everything, so a new language needs *parallel* speech–text data — exactly what low-resource languages lack.
- ▶ In exchange, one network replaces four hand-tuned modules and their error propagation.
- ▶ The field's answers to the data problem are the rest of this lecture: self-supervised pretraining on unlabeled audio (wav2vec 2.0) and massive multilingual supervision (Whisper).

- DNN approximates $P(a \mid x)$ over output tokens; no separate HMM state topology for alignment.



Source: Stanford CS224S, Lecture 10 (2025 Spring), slide 25.

- ▶ Each timestep t the model outputs $P(a \mid X_{1:T})$, which can either be an output token (character, phoneme) or “blank”.
- ▶ *Assume:* Input sequence (length T) $\geq$ output sequence length (safe assumption in ASR).
- ▶ **Collapsing rule:** Ignore all blank tokens, and ignore repeated outputs of the same token.

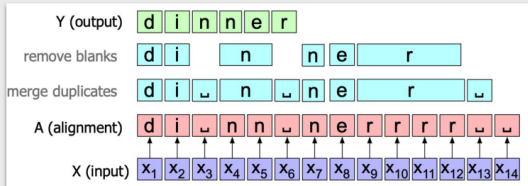


Source: Jurafsky & Martin, *Speech and Language Processing* (3rd ed. draft), ASR chapter.

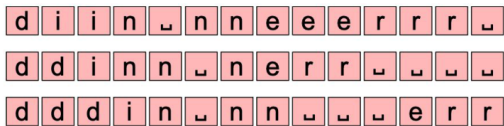
CTC collapsing. Sum over all possible alignments

Many alignments produce the same transcript output.

$P(\text{transcript}) = \text{sum over } P(\text{alignment}) \text{ for all alignments consistent with transcript}$



Additional alignments for 'dinner'



Source: Jurafsky & Martin, *Speech and Language Processing* (3rd ed. draft), ASR chapter: the collapsing function B and further legitimate alignments for *dinner*.

Real example. Full utterance with blank token

Per-frame argmax:

```
yy_ee      tt_      a_
rr_e      hh_      b_ii_    ll_i_    tt_    aa_    tt_    iio_n_
cc_    rrr_u_      ii_    ss_
o_      nn_      hhh_a_      nnddd_      i_n_
thh_e_      bb_uuii_    llldd_    ii_    nng_
l_o_    o_g_g_    ii_    nng_
b_rr_ii_      ck_s_      p_ll_a_      sstt_      eerr_
a_nnd_      b_    ll_uu_      ee_pp_r_    i_    nnss_
f_      oou_      rrr_      f_      oo_rr_tt_y_
t_www_oo_      nn_ew_      b_e_      t_      i_n_
e_      pp_      aa_rr_tt_    mm_ee_    nnntss_
```

After collapsing:

YET A REHABILITATION CRU IS ONHAND IN THE BUILDING LOOGGING BRICKS PLASTER AND BLUEPRINS FOUR FORTY TWO
NEW BETIN EPARTMENTS

Reference

YET A REHABILITATION CREW IS ON HAND IN THE BUILDING LUGGING BRICKS PLASTER AND BLUEPRINTS FOR FORTY TWO
NEW BEDROOM APARTMENTS

Source: Stanford CS224S, Lecture 10 (2025 Spring), slide 28.

Example CTC results. No word lexicon or LM (WSJ corpus)

YET A REHABILITATION CRU IS ONHAND IN THE BUILDING LOOGGING BRICKS PLASTER AND BLUEPRINS FOUR FORTY TWO
NEW BETIN EPARTMENTS

YET A REHABILITATION CREW IS ON HAND IN THE BUILDING LUGGING BRICKS PLASTER AND BLUEPRINTS FOR FORTY TWO
NEW BEDROOM APARTMENTS

THIS PARCLE GUNA COME BACK ON THIS ILAND SOM DAY SOO

THE SPARKLE GONNA COME BACK ON THIS ISLAND SOMEDAY SOON

TRADE REPRESENTIGD JUIDER WARANTS THAT THE U S WONT BACKCOFF ITS PUSH FOR TRADE BARIOR REDUCTIONS

TRADE REPRESENTATIVE YEUTTER WARNS THAT THE U S WONT BACK OFF ITS PUSH FOR TRADE BARRIER REDUCTIONS

TREASURY SECRETARY BAGER AT ROHIE WOS IN AUGGRAL PRESSED FOUR ARISE IN THE VALUE OF KOREAS CURRENCY

TREASURY SECRETARY BAKER AT ROH TAE WOO'S INAUGURAL PRESSED FOR A RISE IN THE VALUE OF KOREAS CURRENCY

Source: Stanford CS224S, Lecture 10 (2025 Spring), slide 29.

- ▶ Labels at each time index are **conditionally independent** given the audio input (like HMMs):

$$\Pr(a \mid x) = \prod_{t=1}^T \Pr(a_t \mid x)$$

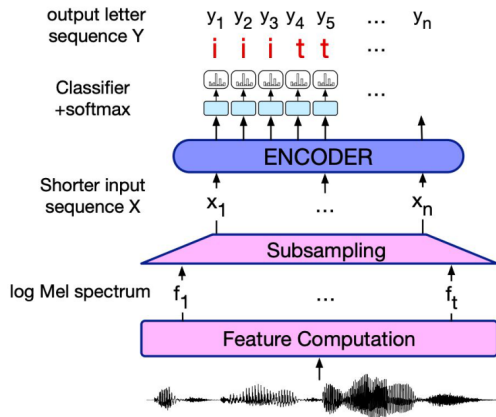
- ▶ **CTC loss:** Negative log probability of the target transcription y^* , summed over all possible time-level labelings a that collapse to y^* :

$$\text{CTC}(x) = -\log \Pr(y^* \mid x), \quad \Pr(y \mid x) = \sum_{a \in B^{-1}(y)} \Pr(a \mid x)$$

- ▶ For sequence length $T = 3$, transcript *HI*:
 - All possible pre-collapsed/time-level labelings: HHI, HII, H_I, _HI, HI_
- ▶ The final loss maximizes the probability of the transcript y^* .
- ▶ **Insights:** Conditional independence allows us to compute outputs for each t in parallel.
 - Blank + collapsing enables efficient, exact summation over all alignments.
- ▶ **But it cuts both ways:** because the outputs are conditionally independent, CTC cannot model dependencies between output tokens. It has no language model of its own, so accurate decoding leans on an **external LM**. This is the gap the RNN-Transducer closes.

CTC inference: Most likely alignment

- ▶ **Output model assumes conditional independence.**
- ▶ Each t most likely output depends only on inputs (not auto-regressive).
- ▶ Most likely pre-collapsed output is simply the sequence of argmax outputs for each t .
- ▶ Run CTC collapsing to obtain transcript.

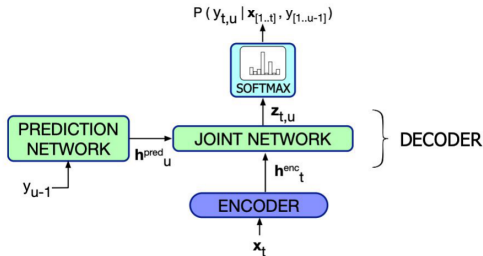


Source: Jurafsky & Martin, *Speech and Language Processing* (3rd ed. draft), Ch. 15 "Automatic Speech Recognition,"

Fig. 15.15.

Improving on CTC: RNN-Transducer (RNN-T) output model

- ▶ **What it fixes:** a **prediction network** conditions on the tokens already emitted, so the model carries its own language model instead of depending on an external one
- ▶ Add network that reasons over sequence of non-blank characters/tokens and input sequence
- ▶ Decoding outputs each transcript token only once. Optional blanks between. Variation on CTC
- ▶ Prediction network often pre-trained with CTC. Encoder provides transformed features for each t



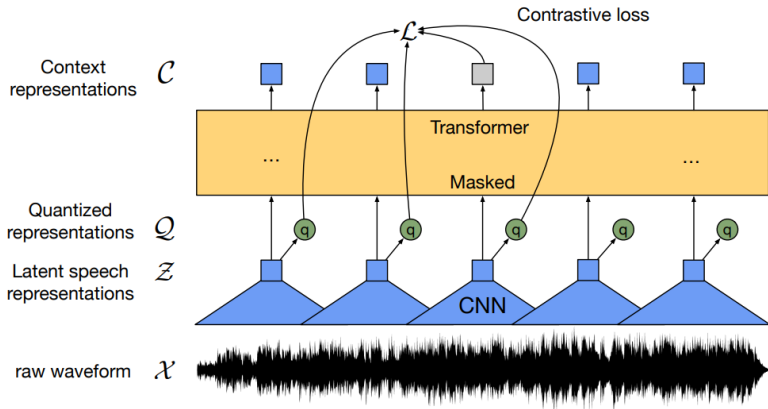
Source: Jurafsky & Martin, *Speech and Language Processing* (3rd ed. draft), ASR chapter.

Self-Supervised Learning: Wav2Vec 2.0

Learning speech representations from unlabeled audio

- ▶ **Proposed by Baevski et al. (2020)**
- ▶ **Key Idea:** Self-supervised learning for speech representation.
- ▶ **Architecture:**
 - CNN encoder for raw waveform $\rightarrow$ latent representations z_t
 - Transformer for context representations c_t
 - Quantization module for discrete representations q_t

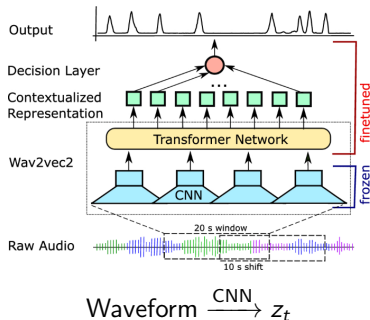
Wav2Vec 2.0: Overview (cont.)



Wav2Vec 2.0 architecture overview.

Source: Baevski et al., wav2vec 2.0, Fig. 1; [arXiv:2006.11477](https://arxiv.org/abs/2006.11477).

- **Step 1:** Raw audio waveform is input to a multi-layer CNN encoder.
- **Output:** Latent speech representations z_t .



Source: Kunešová et al., "Comparison of wav2vec 2.0 models on three speech processing tasks," *Int. J. Speech Technology* 27(4):847–859, 2024, Fig. 2; [doi:10.1007/s10772-024-10140-6](https://doi.org/10.1007/s10772-024-10140-6). Shown there as a fine-tuned frame classifier, not as the wav2vec 2.0 pre-training model.

- ▶ **Step 2:** Random spans of z_t are masked.
- ▶ **Step 3:** Transformer network produces contextualized representations c_t .

$$z_t \xrightarrow{\text{mask spans}} \xrightarrow{\text{Transformer}} c_t$$

- ▶ **Step 4:** Quantize z_t to discrete representations q_t .
- ▶ **Quantization:** Product of G codebooks, each of size V .

$$z_t \xrightarrow{\text{Quantization}} q_t$$

- **Contrastive loss.** At each masked step, identify the true quantized latent q_t within a candidate set Q_t holding q_t itself plus $K = 100$ distractors sampled from other masked steps of the same utterance.

$$\mathcal{L}_m = -\log \frac{\exp(\text{sim}(c_t, q_t)/\kappa)}{\sum_{\tilde{q} \in Q_t} \exp(\text{sim}(c_t, \tilde{q})/\kappa)}$$

- sim is cosine similarity and $\kappa = 0.1$ is a temperature. The sum runs over *all* of Q_t , the positive included, so the ratio is a softmax over candidates.
- **Diversity loss** $\mathcal{L}_d$ encourages use of every codebook entry, stopping the quantizer collapsing onto a handful of codes. Total objective: $\mathcal{L} = \mathcal{L}_m + \alpha \mathcal{L}_d$.

Source: Baevski et al., wav2vec 2.0, Eq. 3; [arXiv:2006.11477](https://arxiv.org/abs/2006.11477).

- ▶ LibriSpeech **test-clean** WER, LARGE model pre-trained on 53.2k hours of unlabeled LibriVox audio and decoded with a Transformer LM:
 - 1.8% fine-tuned on the full 960 h of labeled data (3.3% on test-other)
 - 4.8% fine-tuned on only 10 minutes of labeled data (8.2% on test-other)
- ▶ **The language model is doing real work.** Strip it out and that same 10-minute model lands near 40% WER. Always quote self-supervised numbers with the decoding setup attached.
- ▶ **Impact:** makes unlabeled audio usable, which matters most where labeled speech is scarce.

TIMIT phoneme recognition accuracy in terms of phoneme error rate (PER).

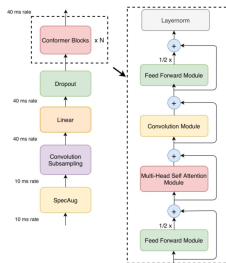
	dev PER	test PER
CNN + TD-filterbanks [59]	15.6	18.0
PASE+ [47]	-	17.2
Li-GRU + fMLLR [46]	-	14.9
wav2vec [49]	12.9	14.7
vq-wav2vec [5]	9.6	11.6
This work (no LM)		
LARGE (LS-960)	7.4	8.3

Source: Baevski et al., wav2vec 2.0: the WERs above are LibriSpeech (Tables 1, 2 and 9); the table shown is Table 3, a *separate* TIMIT phoneme-error-rate benchmark decoded without an LM. [arXiv:2006.11477](https://arxiv.org/abs/2006.11477).

Modern ASR

Conformer encoders and Whisper-scale weak supervision
define current practice

- ▶ Sequence-to-sequence transformer with multi-headed self attention.
- ▶ Combines attention (global context) with convolution (local invariance).
- ▶ Uses RNN-T loss architecture.

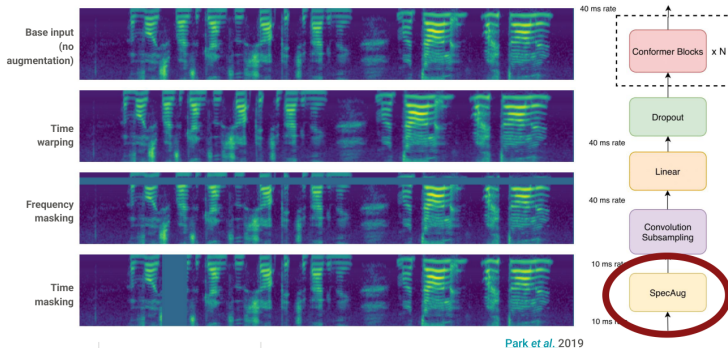


Conformer encoder block: two macaron-like feed-forward layers with half-step residual connections sandwich the multi-headed self-attention and convolution modules, followed by a post layernorm.

Source: Gulati et al., "Conformer," Fig. 1; [arXiv:2005.08100](https://arxiv.org/abs/2005.08100).

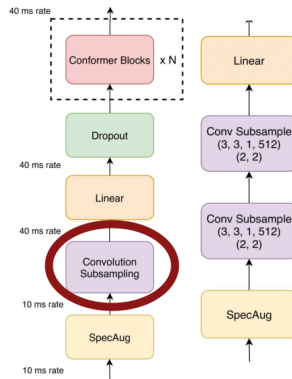
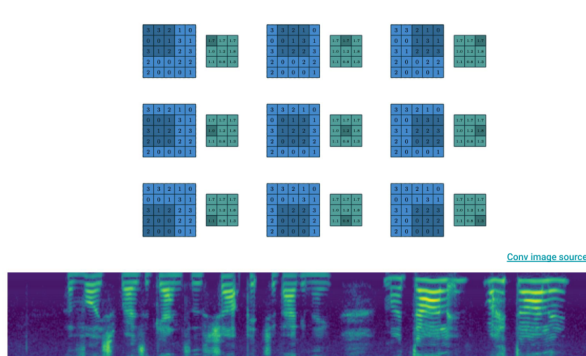
SpecAugment: Augmenting data to increase dataset size

- ▶ Three operations on the log-mel input during training: **time warping**, **frequency masking**, and **time masking**.
- ▶ Applied on the fly at the front of the encoder, so every epoch sees a different view of the same utterance.



Source: Spectrograms: Park et al., “SpecAugment,” Fig. 1; [arXiv:1904.08779](https://arxiv.org/abs/1904.08779). Encoder pipeline: Gulati et al., Conformer, Fig. 1. Via Stanford CS224S, Lecture 11 (2025 Spring), slide 5.

Convolution subsampling: Log mel spectrogram is an image



Source: Grid animation: Dumoulin & Visin, “A guide to convolution arithmetic for deep learning,” Fig. 1.5, which is a 3×3 average pooling over a 5×5 input rather than a convolution; [arXiv:1603.07285](https://arxiv.org/abs/1603.07285). Encoder column: Gulati et al., Conformer, Fig. 1 (left panel); the conv-subsample expansion at right is CS224S’s own. Via Stanford CS224S, Lecture 11 (2025 Spring), slide 6.

- ▶ From ablation experiment of the paper, **removing unique features and moving towards a vanilla Transformer block** was tested.
- ▶ Each row removes one more Conformer-specific component, cumulatively.
- ▶ All ablation study results are evaluated without external LM.

Model Architecture	dev clean	dev other	test clean	test other
Conformer Model	1.9	4.4	2.1	4.3
– SWISH + ReLU	1.9	4.4	2.0	4.5
– Convolution Block	2.1	4.8	2.1	4.9
– Macaron FFN	2.1	5.1	2.1	5.0
– Relative Pos. Emb.	2.3	5.8	2.4	5.6

Source: Gulati et al., “Conformer,” Table 3;

[arXiv:2005.08100](https://arxiv.org/abs/2005.08100).

Multitask training data (680k hours)

English transcription

 "Ask not what your country can do for ..."

 Ask not what your country can do for ...

Any-to-English speech translation

 "El rápido zorro marrón salta sobre ..."

 The quick brown fox jumps over ...

Non-English transcription

 "언덕 위에 올라 내려다보면 너무나 넓고 넓은 ..."

 언덕 위에 올라 내려다보면 너무나 넓고 넓은 ...

No speech

 (background music playing)

(Radford et al. 2022)



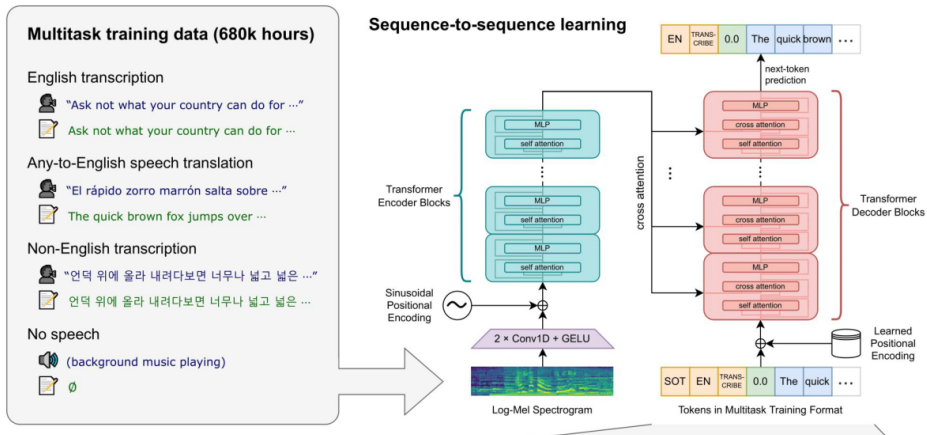
Ground truth: Comme moi, vous avez peut-être déjà vécu cette scène. Qu'est-ce que tu fais cet été toi? Écoute, je ne sais pas encore trop.

Whisper (French): Comme moi, vous avez peut-être déjà vécu cette scène. Qu'est-ce que tu fais cet été toi? Écoute, je ne sais pas encore trop.

Whisper (Translation): Like me, you may have already experienced this scene. What are you doing this summer? I don't know yet

Source: Training-data panel: Radford et al., Whisper, Fig. 1; [arXiv:2212.04356](https://arxiv.org/abs/2212.04356). Video still: France 2, *Envoyé Spécial*. Via Stanford CS224S, Lecture 11 (2025 Spring), slide 18.

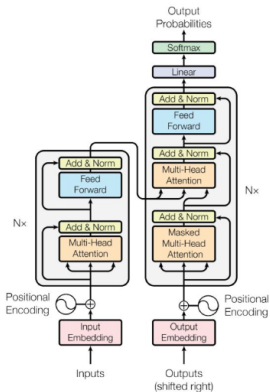
Whisper: How does it perform a variety of tasks accurately?



(Radford et al. 2022)

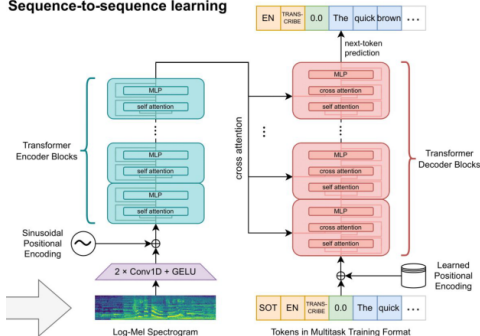
Source: Radford et al., Whisper, Fig. 1 ("Overview of our approach"); [arXiv:2212.04356](https://arxiv.org/abs/2212.04356).

Text encoder-decoder transformer



Whisper encoder-decoder transformer

Sequence-to-sequence learning

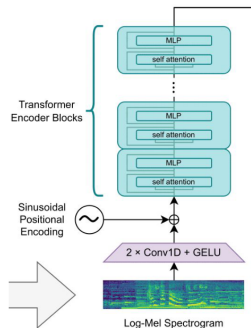


Source: Left: Vaswani et al., "Attention Is All You Need," Fig. 1; [arXiv:1706.03762](https://arxiv.org/abs/1706.03762). Right: Radford et al., Whisper, Fig. 1;

[arXiv:2212.04356](https://arxiv.org/abs/2212.04356).

Whisper encoder: transformer encoder

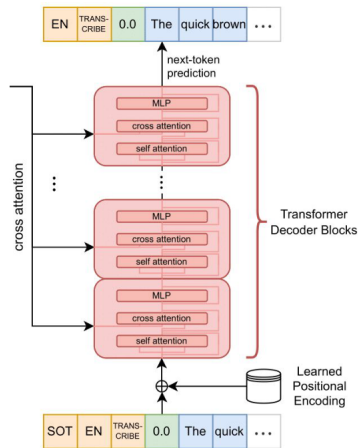
- ▶ Encoder input: log-mel spectrogram and two convolutional layers
- ▶ Sinusoidal positional encoding
- ▶ Standard transformer blocks depending on model size



Source: Radford et al., Whisper, Fig. 1 (encoder panel);

[arXiv:2212.04356](https://arxiv.org/abs/2212.04356).

- ▶ **GPT-2 style** model: no more CTC loss.
- ▶ Decodes using **auto-regressive prediction**: each token depends on previous (text prediction).
- ▶ **Cross-attention**: To encoded audio representation from encoder.
- ▶ **Custom vocabulary and tags**:
 - Language tags: <en>, <fr>, etc.
 - Task tags: <transcribe>, <translate>
 - **Timestamps**
 - Speech tags: <nospeech>

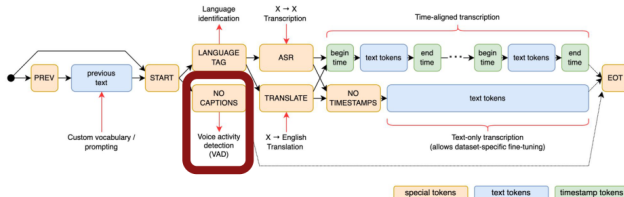


Source: Radford et al., Whisper, Fig. 1 (decoder panel);

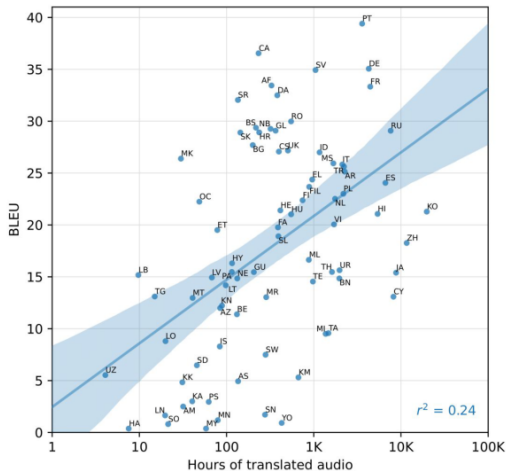
[arXiv:2212.04356](https://arxiv.org/abs/2212.04356).

Whisper handles long-form transcription

- ▶ Whisper only takes 30 seconds of speech at a time
- ▶ **Timestamp tokens** are used to determine when to shift the window
- ▶ Text from the previous window is provided with transcribing current window
- ▶ Voice activity detection using the `<nospeech>` tag



Source: Stanford CS224S, Lecture 11 (2025 Spring), slide 30: a redraw of Radford et al., Whisper, Fig. 1 (multitask training format), with the VAD branch ringed in red by the lecturer. The published figure reads `START OF TRANSCRIPT, NO SPEECH` and `TRANSCRIBE`; [arXiv:2212.04356](https://arxiv.org/abs/2212.04356).



Speech translation performance on FLEURS dataset (higher is better)

- ▶ Translation performance is better on average after the 100 hours of training data mark.
- ▶ Amount of training data isn't as correlated with translation performance as it is with transcription performance.
- ▶ Although possible, speech translation is likely to only be reliable for high resource languages.

Source: Radford et al., Whisper, Fig. 4; [arXiv:2212.04356](https://arxiv.org/abs/2212.04356).

- ▶ Scale steadily reduced **perception-level** errors such as confusing similar-sounding words. What remains the paper calls “decidedly non-human/perceptual”, and it bites hardest in long-form transcription.
- ▶ Three named failure modes, arising from the interaction of seq2seq decoding, the language model, and text-audio alignment:
 - getting stuck in **repeat loops**
 - failing to transcribe the **first or last few words** of a segment
 - **complete hallucination**, where the output is a transcript entirely unrelated to the audio
- ▶ These outputs are fluent, so they do not look like errors. Anything downstream that trusts the transcript inherits them silently.
- ▶ Coverage is uneven: the pre-training mix is English-heavy and most languages have under 1000 hours, so per-language accuracy tracks per-language data volume.

Source: Radford et al., Whisper, §6 “Limitations and Future Work”; [arXiv:2212.04356](https://arxiv.org/abs/2212.04356).

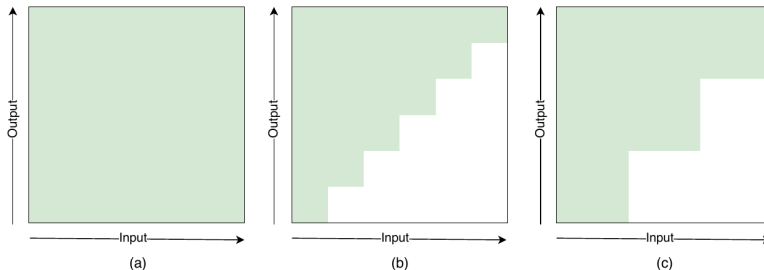
Streaming ASR

Transcribing while the user is still speaking

- ▶ **Offline (full-context):** the whole utterance is available before decoding starts. Best accuracy; fine for transcribing recordings.
- ▶ **Streaming:** tokens must appear while audio is still arriving. Required for voice assistants, live captions, and dictation.
- ▶ The core constraint: a streaming model may only use the **past** (and a small fixed lookahead), never the whole future of the utterance.
- ▶ User-perceived latency comes from three places: acoustic lookahead, model computation, and the decision that the speaker has finished.

- ▶ Full self-attention encoders (Conformer, Whisper) attend over the entire utterance, so they cannot stream as-is.
- ▶ **Causal encoders:** restrict attention and convolutions to the past only. Cheapest latency, largest accuracy loss.
- ▶ **Chunked attention:** process audio in fixed chunks with a small right-context window; each chunk sees a little of the future.
- ▶ Chunk size is the latency-accuracy knob: bigger chunks recover offline accuracy, smaller chunks respond faster.

What the encoder is allowed to see



Green marks the input positions each output position may attend to: (a) full attention, the offline case; (b) left-only attention, fully causal; (c) chunk attention, where every chunk sees itself and all previous chunks.

Source: Zhang et al., "Unified Streaming and Non-streaming Two-pass End-to-end Model for Speech Recognition," Fig. 2;

[arXiv:2012.05481](https://arxiv.org/abs/2012.05481).

- ▶ The RNN-Transducer is **frame-synchronous**: at each audio frame it either emits a token or a blank and moves on, so partial transcripts appear as audio arrives.
- ▶ No attention over the full input and no need to wait for the end of the utterance, unlike encoder-decoder models with cross-attention.
- ▶ The prediction network conditions on previously emitted tokens, giving language-model-like context without breaking streamability.
- ▶ Production systems (mobile dictation, on-device assistants) are overwhelmingly RNN-T variants with causal or chunked encoders.

- ▶ **Endpointing** is deciding that the speaker has finished, so the final result can be committed and the microphone closed.
- ▶ Too eager: the system cuts users off mid-sentence. Too patient: every interaction feels slow.
- ▶ Signals used: silence duration, the decoder's end-of-utterance probability, and prosodic cues.
- ▶ Endpointing errors often dominate perceived quality even when word accuracy is high.

► How many hours of data do you have?

- If you have hundreds of hours of data and need speech recognition for a specific domain, you could get away with training your own model or fine tuning an existing model.
- If you have tens of hours of data, you can finetune an existing model.

► Which language are you looking to transcribe or translate?

- If the model is well represented in Whisper, it is a strong baseline, especially for translation into English.
- If you'd like to translate into another language, you will need your own model.

► What is your compute budget?

- If you don't have any compute budget, using the best model you can find may be the best way forward.
- If you have a small compute budget, finetuning a small model variant could outperform a larger model out-of-the-box.
- If you have a large compute budget (and large enough dataset) you can consider finetuning a larger model variant.

- ▶ Noise, codecs, mic quality, overlap, and domain mismatch dominate field WER.
- ▶ **Demographic bias is measured, not hypothetical.** Across five commercial systems (Amazon, Apple, Google, IBM, Microsoft), average WER was **0.35** for Black speakers against **0.19** for white speakers on age- and gender-matched interview audio.
- ▶ The gap held on identical phrases spoken by both groups, which traces it to the acoustic models rather than to language or topic.
- ▶ Audit before deploying in justice, hiring, or grading use cases.
- ▶ Voice is biometric: consent, retention, on-device options, and transparent API error rates.

Source: Koenecke et al., “Racial disparities in automated speech recognition,” *PNAS* 117(14):7684–7689, 2020.

- ▶ **Datasets:** LibriSpeech, GigaSpeech, Common Voice, AMI, CHiME, FLEURS. Match to deployment domain.
- ▶ **Toolkits:** ESPnet, NeMo/Riva, Fairseq, SpeechBrain, Hugging Face transformers.



Source: ESPnet logo from github.com/espnet/espnet; Watanabe et al., “ESPnet: End-to-End Speech Processing Toolkit,” Interspeech 2018; [arXiv:1804.00015](https://arxiv.org/abs/1804.00015).

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